Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Parameter updates can reinforce or cancel one another even when training loss continues to decrease. We study this distinction through an exact finite-time decomposition of displacement under stochastic gradient descent (SGD), separating gradient magnitudes from interactions across all temporal lags. Consecutive-gradient cosine provides a local directional diagnostic, while normalized displacement measures cumulative retention of traveled distance. Local expansions connect these quantities to curvature and sampling noise without identifying alignment sign with stability or objective progress. Full-batch experiments exhibit gradient reversal alongside continued loss reduction. Matched mini-batch continuations show that sampling changes both gradient estimates and the paths traversed through the objective. We then extend the diagnostic interpretation to Adam by distinguishing raw gradients, first moments, adaptive directions and actual parameter increments. Language-model measurements and checkpoint-matched schedule interventions demonstrate why coherence at these different stages must be assessed alongside loss. The resulting framework supports the formulation and evaluation of informed learning-rate interventions; it does not prescribe an alignment-maximizing schedule.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Learning-Rate Scheduling**

I. Introduction

Neural scaling laws have ushered in new paradigms for interpreting deep learning dynamics. By establishing that performance gains follow predictable behavior relative to model size, dataset scale, and compute, stochastic optimization methods have been pushed toward increasingly extended training horizons while operating under tight efficiency constraints [1]–[4]. In this regime, optimization stability becomes an important practical requirement, since unstable training may result in slow convergence and high sensitivity to hyperparameter choices. In practice, stability is often maintained through a set of heuristics, such as momentum, batch-size scaling, normalization, and learning-rate schedules [5]–[9]. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

Thiago Mattar, and Yan G. Ferreira are with the Graduate Program in Electrical Engineering, Universidade Federal de Minas Gerais, Belo Horizonte 31270-901, Brazil (e-mail: thiagocmattar@gmail.com; yanvgferreira@gmail.com).

Cristiano L. Castro is with the Graduate Program in Electrical Engineering and the Department of Electrical Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 31270-901, Brazil (e-mail: crislcastro@gmail.com).

Frederico Coelho, and Antonio P. Braga are with the Graduate Program in Electrical Engineering and the Department of Electronic Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 35931-008, Brazil (e-mail: fredgfc@ufmg.br; apbraga@ufmg.br).

Classical local analyses relate the permissible step size to curvature in sharp directions of the loss [10], [11]. Stochastic extensions, including the break-even point framework, also account for curvature variation induced by sampling [12]. The Edge-of-Stability literature shows that deep networks can continue to make progress while oscillating near, or beyond, the nominal sharpness threshold [13]–[15]. These results motivate a complementary diagnostic question: how do the magnitudes and directions of successive gradients combine into parameter motion along the observed training trajectory? Gradient information already computed during optimization makes this question accessible without estimating the full Hessian spectrum or gradient-noise covariance at every update.

To examine these interactions, we describe parameter motion through gradient magnitudes and directional alignment. Starting from an untrained reference, training changes the parameters through which the model expresses what it has learned. Displacement provides a geometric description of that change; it does not quantify retained knowledge or establish whether the change improves the objective. Positive and negative gradient interactions respectively reinforce and cancel components of the accumulated motion. Their effect must be evaluated at the relevant temporal horizon and distinguished from loss reduction.

We develop an exact finite-time decomposition of SGD displacement that retains all cross-step interactions. Consecutive-gradient cosine supplies a local online diagnostic, while normalized displacement measures the fraction of accumulated step length retained as net motion. Local expansions relate gradient changes to directional curvature and sampling noise. The distinction between these quantities is consequential: a trajectory can reverse successive gradient directions while continuing to reduce loss, and sampling can change both the measured alignment and the subsequent path.

The same distinction becomes essential under adaptive optimization. Adam already computes raw gradients and moment estimates [16]; its parameter updates combine temporal averaging, coordinate scaling and the learning rate. Monitoring these successive quantities can reveal changes that a single alignment statistic would obscure. We use language-model measurements and schedule interventions to illustrate their interpretation, while separating descriptive geometric responses from the development of an improved learning-rate rule.

Our contributions are:

- A displacement-based account of local and cumulative gradient interaction, with explicit connections to curvature, sampling noise and finite-step loss changes;

- Controlled full-batch and mini-batch experiments that distinguish directional reversal, displacement retention and objective progress;
- A practical extension to Adam that separates raw-gradient information, momentum, adaptive directions and actual parameter motion, supported by language-model measurements and matched schedule interventions.

The remainder of the paper is organized as follows. Section 2 reviews training-stability perspectives. Section 3 develops the displacement formulation and its diagnostics. Sections 4 and 5 examine full-batch GD and mini-batch sampling. Section VI extends the interpretation to Adam and outlines the empirical requirements for informed scheduling. Section 7 concludes.

II. Background: Stability Analysis of SGD

Intuitively, stable optimization occurs when updates reinforce each other and gradually shift parameters toward a point of minimum loss. Instability therefore arises when updates overshoot, oscillate, or partially undo the progress made by previous steps [10], [11]. Understanding this behavior requires examining the mechanisms that govern how updates evolve from one iteration to the next. Two factors have traditionally played a central role in explaining these dynamics: the *local curvature* of the loss landscape and the *stochasticity* introduced by mini-batch sampling.

A. Local Curvature of the Loss Function and Stability

Training can be interpreted as a discrete-time dynamical system in which the local curvature of the loss function determines the stability of the updates [10], [11]. The curvature indicates how sensitive the loss is to parameter changes in different directions: when curvature is small, updates tend to produce smooth descent toward a minimum, whereas in directions of high curvature the same step can overshoot the minimum and induce oscillatory behavior [10], [11], [17]. For this reason, classical analyses of gradient descent relate stability to the spectrum of the Hessian of the loss function, whose eigenvalues quantify directional curvature and indicate how sensitive optimization is to the choice of learning rate [10], [11].

Under this view, let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, the SGD update is

$$\theta_t = \theta_{t-1} - \eta\, g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$, where $\mathcal{L}$ denotes the full training objective. Defining $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of the deterministic update map is

$$J_t = I - \eta\, H_t. \tag{2}$$

Consider a small perturbation e_{t-1} around θ_{t-1}. Under the local linearization of the full-batch GD update, this perturbation evolves as

$$e_t \approx J_t e_{t-1} = (I - \eta H_t) e_{t-1},$$

In a locally quadratic regime near a minimum, the Hessian is approximately constant and admits eigenpairs (λ_i, v_i). Decomposing the perturbation as $e_t = \sum_i \delta_t^{(i)} v_i$, each eigendirection evolves independently according to

$$\delta_t^{(i)} = (1 - \eta\lambda_i)\, \delta_{t-1}^{(i)}.$$

Thus, the factor $1 - \eta\lambda_i$ determines whether perturbations along direction v_i decay or grow. A classical necessary and sufficient condition for linear stability is

$$\max_i |1 - \eta\lambda_i| < 1 \;\Leftrightarrow\; \eta\,\lambda_{\max}(H_t) < 2, \tag{3}$$

which means that all perturbation components contract over time [10], [11]. In other words, if $0 < \eta\lambda_i < 1$, the perturbation decays monotonically along v_i; if $1 < \eta\lambda_i < 2$, it alternates sign while shrinking, producing oscillations around the minimum; and if $\eta\lambda_i > 2$, it alternates sign while growing, leading to divergence. In over-parameterized networks, however, the Hessian spectrum is typically highly anisotropic, with a small number of outlier directions carrying most of the sharpness, while the bulk remains near zero [18], [19]. Although this suggests monitoring $\lambda_{\max}(H_t)$ as a stability diagnostic, estimating leading Hessian eigenvalues online in large neural networks is computationally expensive, and mini-batch estimates provide only noisy, pointwise proxies of the local geometry.

B. Mini-Batch Stochasticity

A complementary line of work studies SGD through the lens of stochastic optimization, interpreting the mini-batch gradient as a noisy approximation of the full gradient computed over the entire dataset [20]–[23]. In this sense, each SGD step combines a deterministic descent component with a zero-mean random fluctuation induced by sampling, which covariance provides information about how noise is distributed across parameter-space directions and how it may affect trajectory stability. Formally, under fresh, conditionally unbiased sampling, we write the mini-batch gradient as

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &\approx \frac{1}{b}\,\Sigma(\theta_{t-1}), \end{aligned} \tag{4}$$

where ξ_t is the sampling noise, b is the batch size, and $\Sigma(\theta_{t-1})$ denotes the covariance matrix of per-sample gradients around the full gradient. At the current parameters, write $h_i := \nabla\ell_i(\theta_{t-1})$, where ℓ_i is the loss for sample i, and $\mu := \frac{1}{N}\sum_{i=1}^N h_i = \nabla\mathcal{L}(\theta_{t-1})$, where N is the dataset size. The covariance matrix is

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}\big(h_i - \mu\big)\big(h_i - \mu\big)^\top, \tag{5}$$

and the scalar dispersion of per-sample gradients around the full gradient is given by the trace of $\Sigma(\theta_{t-1})$

$$\mathrm{tr}(\Sigma(\theta_{t-1})) = \frac{1}{N}\sum_{i=1}^{N}\|h_i - \mu\|^2.$$

This makes clear why $\Sigma(\theta_{t-1})$ is generally not a viable online diagnostic: computing it requires access to the full gradient μ and all per-sample gradients $\{h_i\}_{i=1}^N$ at each training step, which is impractical and incurs at least a full-pass cost per step.

C. Curvature–Noise Interaction

The *break-even point* framework of Jastrzebski et al. [12] is an attempt to extend the classical curvature-based stability condition to SGD. The central idea is to analyze stability along the sharpest direction of the loss landscape while explicitly accounting for mini-batch stochasticity. More precisely, the framework proposes a one-dimensional necessary-and-sufficient condition for SGD stability at iteration t,

$$(1 - \eta\lambda_{\max}(H_t))^2 + s_t^2 \frac{\eta^2(N-b)}{b(N-1)} \;\leq\; 1, \tag{6}$$

where $\lambda_{\max}(H_t)$ is the top Hessian eigenvalue, N is the dataset size, b is the batch size, and s_t^2 denotes the variance across training examples of the curvature measured along the leading Hessian direction. Large s_t^2 means different mini-batches see substantially different local curvature, while small s_t^2 means they see similar curvature. In this sense, Eq. (6) can be read as a stochastic extension of the deterministic stability condition in Eq. (3), so that the first term captures curvature-induced amplification, while the second captures the destabilizing effect of mini-batch stochasticity.

Under this definition, the break-even point is defined as the first iteration at which the left-hand side of Eq. (6) reaches 1. Intuitively, it marks the end of an early phase during which the choice of learning rate and batch size can strongly shape the later optimization trajectory. In this sense, a central conjecture of this framework is that, prior to this point, hyperparameters largely determine the maximum curvature and stochasticity levels encountered later in training, thereby setting the regime in which the subsequent dynamics unfold.

Although the local analysis simplifies both the interpretation and the mathematical treatment of the stability problem, it is subject to well-known limitations. For instance, the Edge-of-Stability (EoS) phenomenon challenges the local-quadratic picture even in deterministic gradient descent (GD). Across architectures and tasks, full-batch GD often exhibits progressive sharpening, with the largest eigenvalue of the Hessian, $\lambda_{\max}(H_t)$, rising until it hovers near (and sometimes above) the nominal threshold $2/\eta$ (Eq. 3), while optimization continues to make progress despite short-term non-monotonicity [13]–[15]. Thus, successful training can coexist with local linear instability for extended periods, indicating that sharpness alone does not fully explain and determine trajectory-level behavior.

Together, these results clarify both the value and the limitation of curvature–noise interaction criteria. Quantities such as the leading Hessian eigenvalue $\lambda_{\max}(H_t)$ and the gradient-noise covariance $\Sigma(\theta_{t-1})$ characterize local geometry and stochasticity, but can be expensive to estimate online. Their pointwise values also leave open how successive updates interact and accumulate over a training trajectory. This motivates complementary diagnostics that describe directional reinforcement and cancellation using quantities available during optimization.

D. Practical Stability Analysis in Large-Model Pretraining

Large-model training reports approach stability through observable symptoms and interventions on the training recipe. An instructive example is PaLM: Section 5.1 reports approximately 20 loss spikes during training of the 540B model despite global gradient clipping. The authors recovered by returning to a checkpoint about 100 steps before a spike and skipping 200–500 batches. Replaying those batches from a different parameter state did not reproduce the spikes, suggesting an interaction between data and model state rather than a property of the batches alone [24].

Reports accompanying released model weights provide further operational evidence. During OPT-175B training, loss divergence was associated with failures of the dynamic loss scale and growing final-layer activation norms. Recovery involved reducing the learning rate and restarting from an earlier checkpoint; lowering the gradient-clipping threshold from 1.0 to 0.3 also improved stability early in training [25]. These observations motivate monitoring the numerical state of training alongside the loss.

OLMo 2 makes this diagnostic process explicit by measuring a rolling spike score on training loss and gradient norm, and using controlled ablations to study initialization, normalization, and optimizer settings. Its revised recipe combines global gradient clipping with changes including query–key normalization, reordered normalization, and a logit regularizer (z-loss) [26]. In Llama 3, the authors report a stable pretraining recipe with few loss spikes and no interventions for divergence; they use learning-rate warmup and a staged batch-size schedule, and emphasize FP32 gradient accumulation and reduction for numerical stability [27]. These reports document both failure recovery and recipes that avoid recurrent failures, rather than a universal instability mechanism.

Loss, gradient norm, and activation statistics are practical monitoring signals, but gradient magnitude alone does not describe the relationship between successive directions. A large gradient can reinforce or oppose its predecessor, and a sequence of individually bounded steps can still exhibit cumulative cancellation. We therefore seek a complementary description that can be computed from training gradients and interpreted through their temporal geometry. The following section develops parameter displacement and consecutive-gradient cosine similarity for this purpose, connecting the latter to directional curvature and gradient noise without requiring online Hessian eigendecomposition.

III. Parameter Displacement under Stochastic Gradient Descent

Loss spikes and oscillatory updates motivate practical diagnostics that can be computed from gradients already available during training. We develop these diagnostics by relating parameter displacement, gradient magnitudes and alignment,

curvature and sampling noise. The proposed quantities characterize constructive and destructive regimes, and inform possible interventions without the additional overhead of computing Hessians.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. As before, $\theta_t \in \mathbb{R}^p$ denotes the parameters after update t, and g_t is the mean mini-batch gradient evaluated at θ_{t-1}. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{7}$$

For a model trained from scratch, θ_0 represents the initialized model that has not yet learned from the training data. After T updates, the trained parameters θ_T encode what the model has learned within that training budget. The displacement Δ_T captures the net parameter change produced by training. Its squared norm measures the magnitude of this change in the chosen parameterization; its relation to learned function or predictive performance requires additional evidence. Expanding the squared displacement separates gradient magnitudes from the interactions between different steps

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t.\end{aligned} \tag{8}$$

The terms $\|g_t\|^2$ are nonnegative and determine the squared lengths of individual SGD steps through $\eta^2\|g_t\|^2$. In full-batch GD, a larger gradient norm indicates a steeper local loss and, therefore, a larger first-order decrease along the negative gradient. The cross-step terms $g_u^\top g_t$ describe how these steps interact – positive inner products reinforce the displacement, whereas negative inner products cancel part of it. Large gradient norms can therefore coexist with cancellation along the training trajectory, and destructive interference (i.e. persistent $g_u^\top g_t < 0$) can limit displacement growth.

To isolate directional alignment from gradient magnitudes, define the cosine similarity between any two nonzero gradients and its consecutive-step form

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \quad 1 \le u < t \le T. \tag{9}$$

In this setup, the alignment between consecutive gradients is $\rho_t := \rho_{t-1,t}$, $t = 2, \ldots, T$. Substituting into Eq. (8) expresses the squared displacement entirely through gradient norms and directional alignment

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}.\end{aligned} \tag{10}$$

For fixed gradient norms, positive $\rho_{u,t}$ increases the squared net displacement, negative $\rho_{u,t}$ contributes cancellation, and orthogonal gradients have no pairwise contribution. In practice, ρ_t provides an online diagnostic of the interaction between successive updates: persistent positive or negative values characterize constructive or destructive regimes.

A. Directional Curvature and Consecutive Alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise in Eq. (4), $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$. Suppose the full objective $\mathcal{L}$ is twice continuously differentiable along the preceding step, and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD update yields

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{11}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\begin{aligned}\|g_{t-1}\|\|g_t\|\rho_t \approx \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}.\end{aligned} \tag{12}$$

The right-hand side separates the preceding gradient magnitude, the curvature contribution, and the gradient noise contribution. Positive curvature along the preceding gradient direction reduces its alignment with the next gradient, while the noise difference can reinforce or oppose this effect. To characterize these contributions, we define the directional curvature and normalized noise contribution

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{13}$$

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{14}$$

Here, κ_t measures the curvature projected at the gradient direction, and ε_t describes the mini-batch noise alignment with the same gradient direction. Substituting these into Eq. (12) and dividing by $\|g_{t-1}\|\|g_t\|$ gives, after neglecting the Taylor remainder, the first-order relation

$$\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t). \tag{15}$$

Rearranging Eq. (15) gives

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx \eta\kappa_t - \varepsilon_t. \tag{16}$$

The quantity $\widehat{\kappa}_t$ is computable from consecutive mini-batch gradients without evaluating the Hessian. However, it combines directional curvature with stochastic noise and therefore does not isolate κ_t. Equation (15) also characterizes the sign of consecutive gradient alignment. Neglecting the Taylor remainder, and in the absence of noise ($\varepsilon_t = 0$), consecutive nonzero gradients are positively aligned when $\eta\kappa_t < 1$, orthogonal when $\eta\kappa_t = 1$, and negatively aligned when $\eta\kappa_t > 1$. Differently from Eq. 3, here we focus on the characterization of constructive-destructive regimes.

B. Noise Contribution under Mini-Batch Sampling

Although mini-batch gradients are unbiased estimators of the full gradient, the noise contribution to consecutive gradient alignment need not vanish in expectation. Under fresh, independent mini-batch sampling across updates, the current noise ξ_t has zero expected inner product with the preceding gradient g_{t-1}. However, g_{t-1} already contains its own noise ξ_{t-1}. Consequently, assuming finite second moments,

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{17}$$

Thus, even unbiased sampling noise contributes a systematic nonpositive term to the expected unnormalized interaction between consecutive gradients. Its sign need not be inherited by $\mathbb{E}[\varepsilon_t]$: the denominator in Eq. (14) is a correlated random squared gradient norm.

This contribution is directly related to mini-batch size. Consider a mini-batch of b examples sampled uniformly without replacement from $N > 1$ training examples. Using the per-sample gradient covariance $\Sigma(\theta)$ defined in Eq. (5), the sampling variance satisfies

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{18}$$

At fixed parameters, smaller batches produce larger sampling variance and, consequently, a larger negative expected noise contribution. For full-batch gradients ($b = N$), this contribution vanishes.

Importantly, this does not imply that the total expected inner product $g_{t-1}^\top g_t$ necessarily becomes more negative with smaller batches. The same sampling variance increases the expected squared norm $\|g_{t-1}\|^2$, and these two contributions cancel in Eq. (12). Nevertheless, increasing noise can reduce directional coherence by increasing gradient magnitudes relative to their common full-gradient component.

To isolate the loss of directional coherence due to sampling, consider independent gradient estimates at the same fixed parameters. This also approximates consecutive updates when the full gradient changes little and the dependence induced by parameter motion is negligible. When the gradient norms are sufficiently concentrated, the expected cosine similarity can be approximated by

$$\mathbb{E}[\rho_t] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{19}$$

The ratio in Eq. (19) is an approximation to an expectation of normalized inner products, rather than an exact identity. For independent, identically distributed nonzero gradients at fixed parameters, the exact expected cosine is nonnegative: it equals the squared norm of the mean unit gradient. Thus this comparison describes the attenuation of common directional alignment toward zero; it cannot describe negative expected alignment caused by parameter motion. Along an SGD trajectory, curvature couples successive updates and can induce negative alignment even when sampling noise is large. Equation (19) therefore does not predict how a destructive regime responds to batch size. That response also depends on the evolving full gradient and its interaction with the sampled updates.

C. Normalized Displacement and Cumulative Interference

While the consecutive cosine similarity ρ_t measures local lag-1 directional alignment, it does not capture interactions between more distant updates (lag$\ge$ 2). To measure their cumulative effect, we define the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta\sum_{t=1}^T\|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T\|g_t\|}, \qquad 0 \le R_T \le 1, \tag{20}$$

For constant-step plain SGD, the denominator represents the total distance traveled in parameter space, while the numerator measures the net displacement from initialization. Thus, R_T quantifies the fraction of accumulated step length retained as net displacement.

Using Eq. (10), this ratio satisfies

$$R_T^2 = \frac{\sum_{t=1}^T\|g_t\|^2 + 2\sum_{1\le u<t\le T}\|g_u\|\|g_t\|\rho_{u,t}}{\left(\sum_{t=1}^T\|g_t\|\right)^2}. \tag{21}$$

Unlike ρ_t, which considers only adjacent gradients, R_T incorporates interactions across all temporal lags. Values near one indicate that the updates retain most of their accumulated path length as net displacement. Smaller values indicate greater loss of path-length retention through directional variation, including cancellation; they do not determine the sign of the total cross-step interaction in Eq. (21).

D. Connection with Loss Spikes and Convergence

Constructive and destructive regimes have been characterized on parameter motion. To connect gradient alignment with optimization progress, consider the loss change along the preceding SGD update. A second-order approximation gives

$$\begin{aligned}\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ \approx -\frac{\eta}{2}\|g_{t-1}\|^2\left(1+\rho_t\frac{\|g_t\|}{\|g_{t-1}\|}\right) \\ +\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t).\end{aligned} \tag{22}$$

The complete derivation is provided at Appendix A. For batch gradient descent, the noise contribution vanishes. From Eq. (15), $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t$. Consequently, the local quadratic model predicts a loss decrease when $\eta\kappa_t < 2$, whereas consecutive gradients become negatively aligned when $\eta\kappa_t > 1$. Thus, the interval $1 < \eta\kappa_t < 2$ describes destructive interference without loss increase. Negative alignment alone is therefore insufficient to characterize a loss spike, once it depends too on the relative gradient magnitudes.

Under mini-batch SGD, the noise contribution can overcome the predicted decrease, producing a loss spike even when $\eta\kappa_t < 2$. Such an increase may be transient if subsequent updates recover the preceding loss level.

IV. Empirical Characterization under Full-Batch Gradient Descent

Full-batch gradient descent provides a controlled setting for examining how consecutive gradient directions relate to cumulative parameter motion and objective progress. The sampling-noise term vanishes, and each gradient directly determines

an update. Building on the observation that oscillatory full-GD training can reduce loss over longer timescales [13], we compare learning rates at matched initialization, objective and update budget. We examine local reversal through ρ_t, cumulative motion through R_t and the complete cross-step interaction, and the accuracy of the local loss approximation in Section III-D. The comparisons characterize the nine recorded trajectories.

A. Experimental Design and Measurement

We use three settings motivated by the Edge-of-Stability experiments [13]: a tanh MLP with two 200-unit hidden layers on the first 5,000 CIFAR-10 training images; ResNet-32 on all 50,000 CIFAR-10 training images; and a two-layer, two-head word-level Transformer on WikiText-2, with embedding and feed-forward dimensions of 200. The MLP minimizes half-squared error summed over the ten output coordinates and averaged over examples. ResNet minimizes mean cross-entropy over 50 fixed groups of 1,000 images, with batch normalization computed within each group. The Transformer minimizes token-weighted negative log-likelihood over 2,088,600 prepared training targets. Every update accumulates the complete fixed objective.

Each condition runs for 3,000 updates at a constant learning rate: $\{0.004, 0.02, 0.04\}$ for the MLP, $\{0.025, 0.05, 0.1\}$ for ResNet, and $\{0.05, 0.1, 0.2\}$ for the Transformer. Rates were selected from earlier training calibrations to expose contrasting dynamics within this horizon, without test-loss selection. Within each case, all rates share one frozen initialization (seed 0) and identical data preparation. There is no momentum, weight decay, gradient clipping, schedule, augmentation or dropout. These are paired comparisons conditional on one initialization per case.

We record full-objective loss and raw-gradient geometry at every update. Gradient g_t is evaluated at θ_{t-1}; loss and displacement at index t refer to θ_t. The first adjacent pair is undefined and omitted; all measured gradient norms are nonzero. Both R_t and the displacement norm in Table I use the gradient-sum identity, whose agreement with stored-parameter displacement is subject to FP32 update roundoff. Smoothed curves use complete 50-observation trailing arithmetic means, computed before logarithmic display. Late means average the last 1,000 defined observations. Preparation, precision, accumulation and spectral measurement details appear in Appendix C.

B. Directional Reversal with Continued Loss Reduction

Figure 1 shows that all nine trajectories reduce their training objective over the recorded horizon. Within each case, the largest selected learning rate attains the lowest final training loss. This comparison is at a matched update budget and does not establish an optimal rate outside the tested set.

The alignment traces in Fig. 2(a)–(c) separate this progress from directional coherence. The MLP at $\eta = 0.004$ and Transformer at $\eta = 0.05$ retain positive adjacent alignment throughout the window. Higher MLP and Transformer rates develop persistent negative alignment, as do all three ResNet rates. For ResNet, the first negative pair occurs at updates 1,121, 494 and 231 as the rate increases. Such a first sign event need not mark a sustained regime: the MLP at $\eta = 0.04$ has an initial reversal followed by renewed positive alignment before its later negative phase. The unsmoothed traces in Appendix C retain these transients.

The late mean cosine is below -0.96 for all seven conditions with late reversal, even though their loss continues to decrease over longer timescales. The MLP illustrates this distinction directly: final losses are 0.3365, 0.2114 and 0.1539 at increasing rates, while the corresponding late mean cosines are approximately 1, -0.9913 and -0.9799. These observations illustrate the distinction in Section III-D between adjacent reversal and longer-timescale objective progress. The accuracy of its one-step loss approximation is examined separately below.

C. Cumulative Geometry and Gradient Magnitude

The lower row of Fig. 2 shows that R_{3000} decreases with learning rate within each case. Absolute displacement nevertheless increases, as reported in Table I. For the MLP, the gradient-sum displacement norm rises from 0.951 to 3.139 and 4.763, while the accumulated path length rises from 1.198 to 64.361 and 185.542. The reduction in R_{3000} therefore reflects faster growth of traveled distance than of net displacement.

The temporal decomposition further separates adjacent reversal from whole-trajectory interaction. At $T = 3000$, the cumulative adjacent term $2\sum_{t=2}^{T} g_{t-1}^{\top} g_t$ is negative in seven conditions, whereas the complete term $2\sum_{1\le u<t\le T} g_u^{\top} g_t$ is positive in all nine. Positive contributions beyond lag one outweigh the adjacent negative contribution in those seven trajectories (Table IV). Thus, negative adjacent alignment and low path-length retention coexist with positive total cross-step interaction over the complete recorded horizon. Neither ρ_t nor R_t alone specifies the sign of that total interaction.

Figure 3(a)–(c) shows that gradient magnitude adds another part of the description. In the MLP and ResNet cases, the later negative-alignment phases accompany increased gradient norms, whereas the lowest-rate MLP norm continues to decrease. In the Transformer, the highest-rate norm remains relatively large while its objective continues downward. These deterministic fluctuations cannot be attributed to mini-batch sampling. Magnitude, alignment and cumulative accumulation therefore describe different aspects of the observed motion.

D. Projected Gradient Change and the Local Loss Approximation

The projected change in Fig. 3(d)–(f) approaches two in the late reversing regimes, with late means between 1.9707 and 1.9937. This is consistent with nearly opposing gradients of comparable norm. Its relationship to the cosine is algebraic: for nonzero gradients, $\eta\widehat{\kappa}_t > 1$ if and only if $\rho_t < 0$. The diagnostics are algebraically linked: $\eta\widehat{\kappa}_t$ also incorporates the consecutive norm ratio, but its threshold crossing does not independently validate a curvature estimate.

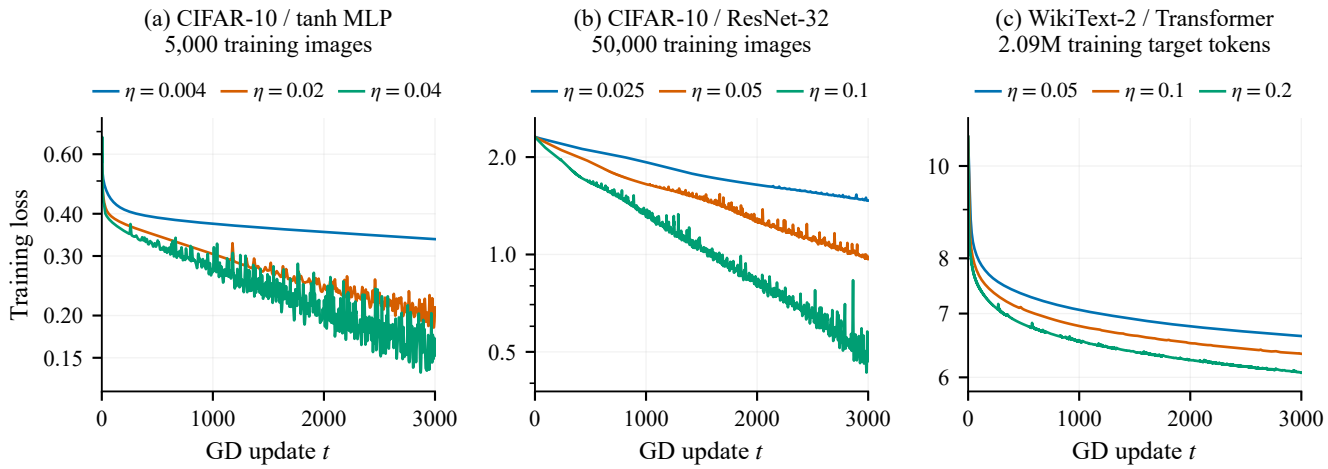


Fig. 1: Unsmoothed full-objective training loss over 3,000 full-batch GD updates, with logarithmic vertical axes. Columns show (a) the tanh MLP, (b) ResNet-32 and (c) the Transformer. Blue, orange and green denote increasing learning rates within each case; the legends give their values. Each case uses one initialization shared across rates (seed 0). Objective values and axis limits differ across tasks and should be compared within columns.

TABLE I: Descriptive endpoints of the nine matched learning-rate conditions. Late means average the last 1,000 defined gradient observations; other entries refer to update 3,000. Displacement is computed as $\eta\|\sum_{t=1}^{3000} g_t\|$, subject to FP32 parameter-update roundoff. Each model case shares one initialization across rates. Values are compared within cases; no across-initialization uncertainty is estimated.

Model	η	$\mathcal{L}(\theta_{3000})$	$\overline{\rho}_{\text{late}}$	R_{3000}	$\overline{\eta\widehat{\kappa}}_{\text{late}}$	$\|\Delta_{3000}\|$
MLP	0.004	0.3365	1.0000	0.7941	4.37×10^{-5}	0.9511
MLP	0.02	0.2114	-0.9913	0.0488	1.9924	3.1391
MLP	0.04	0.1539	-0.9799	0.0257	1.9820	4.7628
ResNet-32	0.025	1.4665	-0.9723	0.1385	1.9740	5.7045
ResNet-32	0.05	0.9672	-0.9708	0.0616	1.9746	9.0183
ResNet-32	0.1	0.4898	-0.9684	0.0374	1.9707	13.9329
Transformer	0.05	6.6301	1.0000	0.7223	1.96×10^{-4}	11.3183
Transformer	0.1	6.3519	-0.9749	0.1395	1.9754	15.6026
Transformer	0.2	6.0697	-0.9937	0.0541	1.9937	21.9693

We directly assess the local loss approximation by comparing

$$\widehat{\delta\mathcal{L}}_{t-1} := -\frac{\eta}{2}\|g_{t-1}\|^2\,(2-\eta\widehat{\kappa}_t) \tag{23}$$

with the recorded change $\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$ for $t = 2,\ldots,3000$. This gives 2,999 unsmoothed comparisons per condition; the last update has no subsequent recorded gradient. The comparison is retrospective because $\widehat{\kappa}_t$ uses that subsequent gradient.

The approximation's sign disagrees with the recorded change in zero transitions for the lowest MLP and Transformer rates, but in 1,325 of 2,999 transitions (44.2%) for the Transformer at $\eta = 0.2$.

After excluding transitions for which either change is at most 10^{-5} times the larger neighboring loss in absolute value, disagreement remains in 1,239 of 2,837 transitions (43.7%). Table V reports every condition, residual magnitudes and threshold sensitivity. These descriptive thresholds do not establish floating-point error bounds. Near a small predicted decrease, a residual can reverse the sign even when it is small relative to the first-order descent term. The audit measures the approximation error without assigning it uniquely to finite-step, nonsmooth or numerical effects.

The nine trajectories therefore demonstrate local reversal alongside positive final all-pair interaction and continued longer-timescale loss reduction. Their gradient-derived diagnostics characterize observed motion; the local loss approximation varies in accuracy across these conditions. The spectral measurements in Appendix C provide additional context with explicit objective coverage. The following section examines sampling noise; transformed optimizer directions and prospective intervention require separate tests.

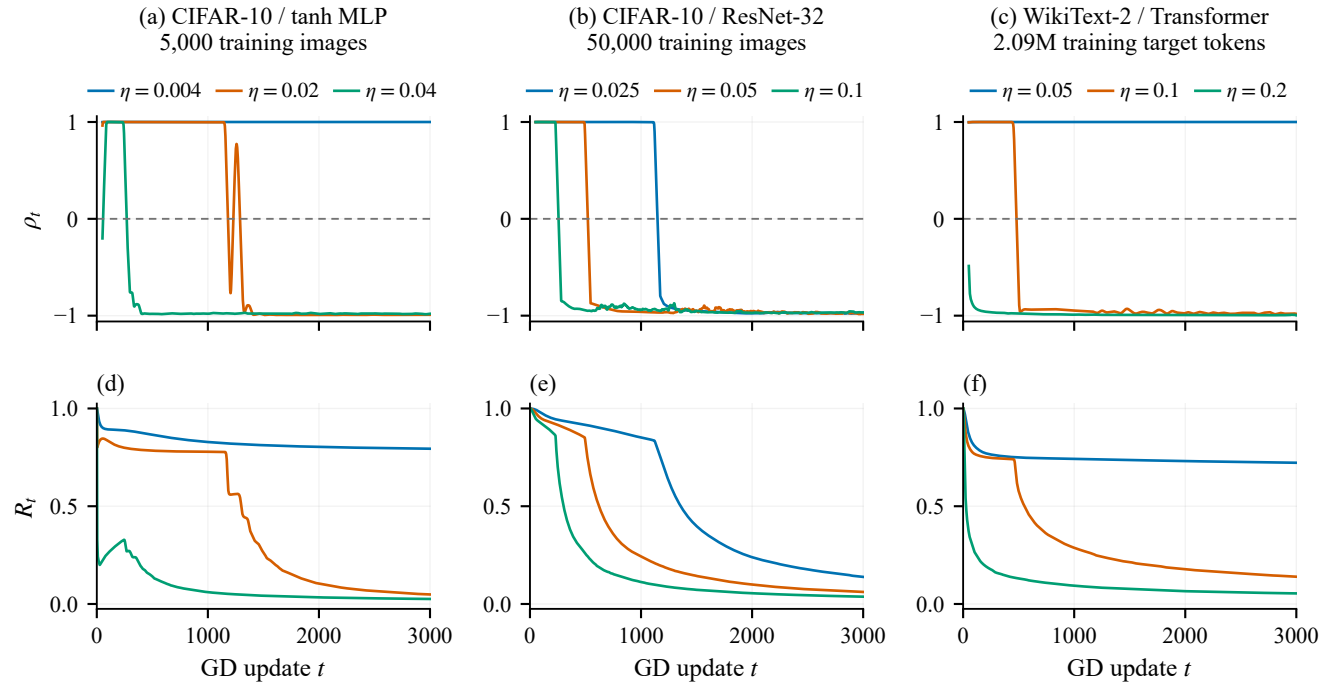


Fig. 2: Local alignment and cumulative displacement in the same nine conditions as Fig. 1. Panels (a)–(c) show consecutive cosine with complete 50-pair trailing means; the dashed reference is zero. Panels (d)–(f) show unsmoothed complete-prefix normalized displacement R_t. Columns are MLP, ResNet-32 and Transformer, with the same LR colors as Fig. 1. Gradients g_t are evaluated before update t; R_t includes that update. The one-initialization curves are descriptive, with no seed averaging.

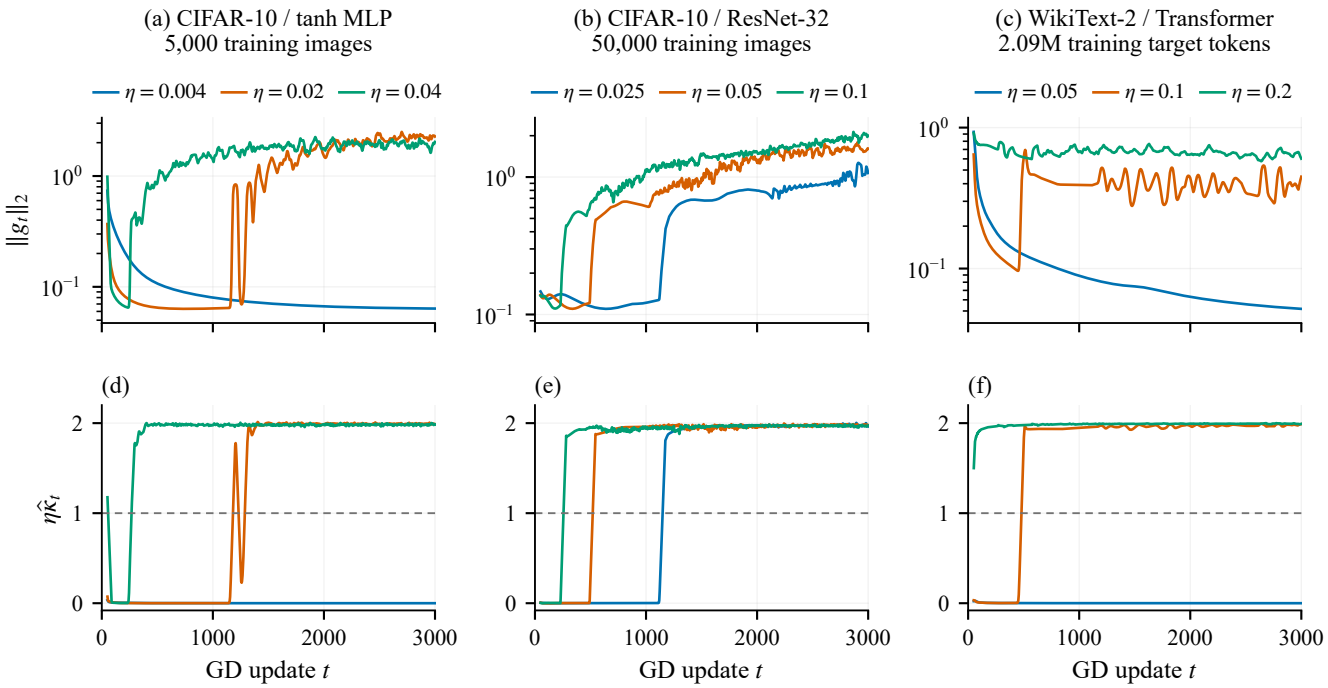


Fig. 3: Gradient magnitude and gradient-derived projected change for the same conditions. Panels (a)–(c) show arithmetic means of raw-gradient norms over complete 50-gradient trailing windows, on logarithmic axes. Panels (d)–(f) show complete 50-pair trailing means of the pointwise $\eta\widehat{\kappa}_t$ from Eq. (16), with a dashed reference at one. Averaging the observable is not equivalent to computing it from averaged gradients or norms. Columns and LR colors follow Fig. 1; no seed aggregation is performed.

V. Effect of Mini-Batch Sampling on the Diagnostics

Mini-batch sampling changes both the estimated gradients and the parameter trajectory. We distinguish these effects by first calibrating the fixed-parameter approximation in Section III-B, then comparing alignment, displacement and loss during matched SGD continuations. The question is how batch size changes the interpretation of the diagnostics at a fixed learning rate.

A. Experimental Design

We use the CIFAR-10 MLP and squared-error objective from Section IV, on the same fixed subset of $N = 5{,}000$ training images. For each of three independent initializations, a new full-batch GD run at $\eta = 0.02$ supplies states before training and after 1,000 and 2,000 updates. Each state initializes 200-update continuations with $b \in \{32, 128, 512, 2048, 5000\}$: eight sampling repetitions per stochastic batch size and one deterministic full-batch control. These are matched starting parameters, not necessarily matched dynamical regimes across initializations. The learning rate remains 0.02 throughout; it is not swept. There is no momentum, weight decay, clipping, schedule, augmentation or dropout.

Each batch is a uniform subset without replacement, independently redrawn at every update. Within a repetition, batch sizes share prefixes of the same random permutation; permutations are independent across updates and repetitions. Parameters and gradients are FP32, diagnostic reductions are FP64, and TF32 is disabled. Full-gradient and loss measurements cover all 5,000 images. The displacement origin is reset at each continuation, so R_{200} measures retention over those 200 updates. Equal update counts imply unequal data exposure.

The calibration instead holds each saved state fixed and draws 32 independent sequences of 128 gradients per batch size. Conditional 95% sampling intervals use 4,000 bootstrap resamples of whole sequences. In the SGD figures, repetitions are averaged within each initialization before the three initializations receive equal weight; shading shows the range of their means, not confidence intervals. Temporal averages describe the stated window without assuming stationarity.

B. Fixed-Parameter Calibration

Using the full gradient and per-example covariance over all 5,000 images, we evaluate Eqs. (17)–(19) without fitted constants. The sampled variance and noise contribution agree with the exact moment identities within 0.50% and 0.74% relative error, respectively. These are implementation checks. The cosine approximation is a separate comparison: it has maximum absolute error 0.0075 and falls outside the conditional sampling interval in 21 of the 36 stochastic conditions. It is accurate in magnitude here, but the deviations are detectable.

At batch 2,048, measured mean cosine falls from 0.986 at initialization to 0.069 at the 1,000-update state; at the latter state, expected squared noise norm is 13.37–13.57 times squared full-gradient norm across initializations. Thus, a large batch fraction can still leave substantial relative noise. As established in Section III-B, independent draws at fixed parameters have nonnegative expected cosine. This calibration cannot test negative alignment during training; that requires parameter motion.

C. Sampling and Parameter Motion

Figure 4 compares consecutive-gradient alignment across batch sizes. The orange curves use full-data gradients evaluated at the same states visited by each SGD continuation; these gradients are measured but do not drive its updates. The blue–orange difference compares two gradient measurements along one path. Comparing orange across batch sizes also involves different parameter paths, with full GD represented only by the endpoint $b = 5000$. The two comparisons are not an additive decomposition of noise effects.

From initialization, batch 32 gives mean sampled cosine -0.141, while full gradients along those trajectories average $+0.353$; the separate full-GD continuation averages 0.994. Sampling therefore changes both the observed alignment and the gradient directions encountered during training. A smaller blue–orange gap does not imply recovery of full-GD behavior: from the 1,000-update state, batch 2,048 gives -0.617 and -0.630, respectively, whereas full GD gives $+0.839$.

Figure 5 resolves this latter comparison in time. At batch 2,048, full-gradient cosine first becomes negative within 3–5 continuation updates in every repetition, and its mean over updates 2–20 is -0.355. In contrast, the full-GD references first become negative at total updates 1,149, 1,227 and 1,224. The 1,000–1,200 observation window therefore includes that transition in only one initialization; the other two retain mean cosine approximately one. This explains why the full-batch endpoint has a positive mean while the largest measured stochastic batch has a negative mean. The contrast reflects changed trajectories and the observation window, not merely attenuated measurement of a common trajectory. Five batch sizes do not resolve a discontinuity or a critical batch threshold.

D. Displacement and Loss

From the 2,000-update state, reducing the batch from 5,000 to 32 moves mean cosine from -0.991 to -0.384. Yet R_{200} remains approximately 0.03 (0.0319 and 0.0303), so weaker consecutive reversal does not yield greater retention of accumulated step length. Full GD reduces mean loss by 0.0087, whereas batch 32 increases it by 0.0241; the increase occurs in every initialization-specific mean. Destructive alignment is therefore compatible with objective progress, and its attenuation is compatible with a worse endpoint loss.

The displacement response also depends on the starting state. Increasing batch size raises mean R_{200} from 0.069 to 0.807 at initialization and from 0.035 to 0.748 at the 1,000-update state, while the late-state values remain near 0.03. Loss improvement increases with batch size in the first two comparisons, but is not monotonic in the third: batch 2,048 attains lower final loss than full GD in all three initializations.

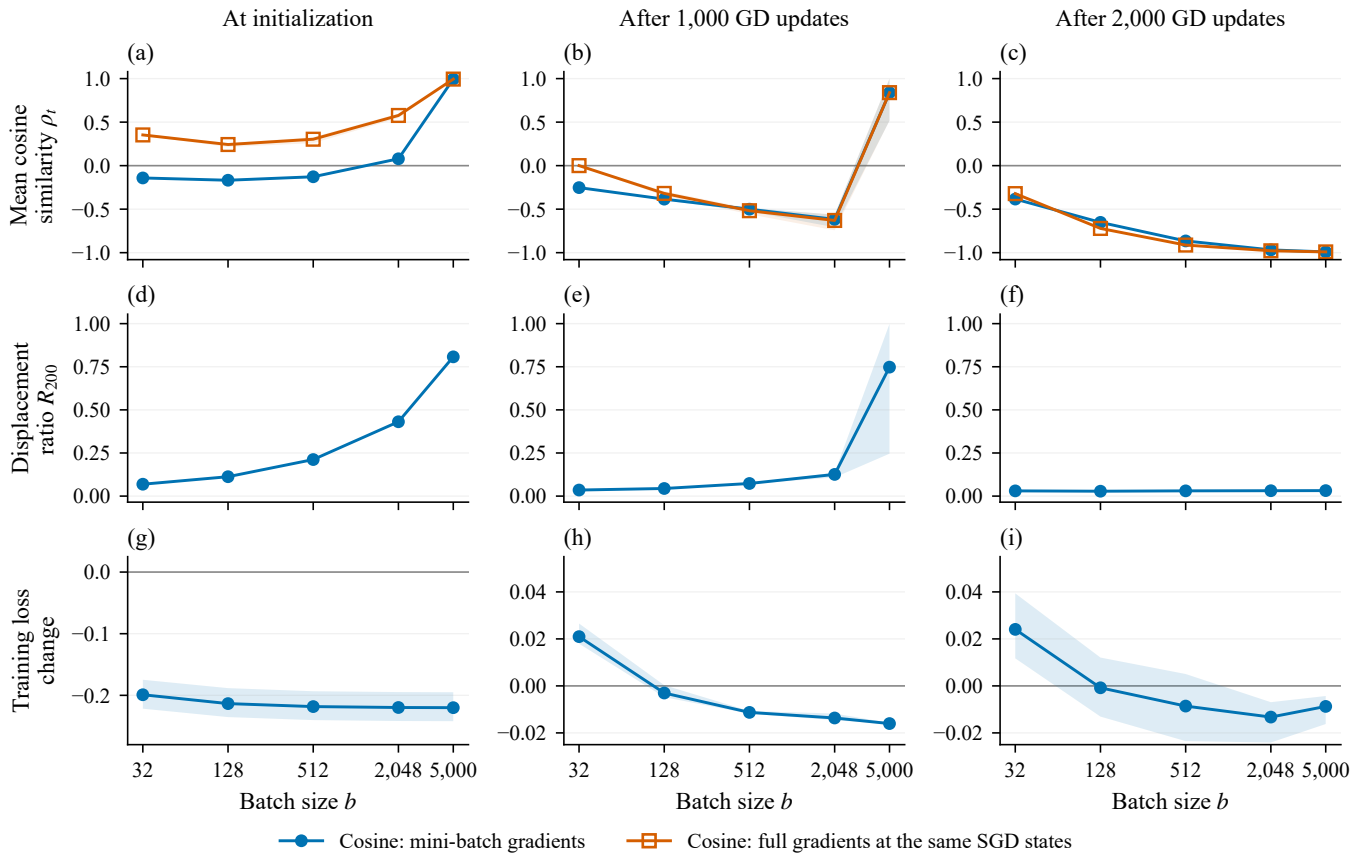


Fig. 4: Batch-size intervention over 200 updates from each saved GD state. Top: mean cosine of sampled gradients (blue) and full gradients at the same SGD states (orange). Only $b = 5000$ follows a full-GD trajectory. Middle: normalized displacement R_{200}. Bottom: final minus initial full-data training loss; positive values indicate an increase. Shading is the range of three initialization means after averaging repetitions. Batch axes are logarithmic and lines guide the eye; loss axes use different ranges.

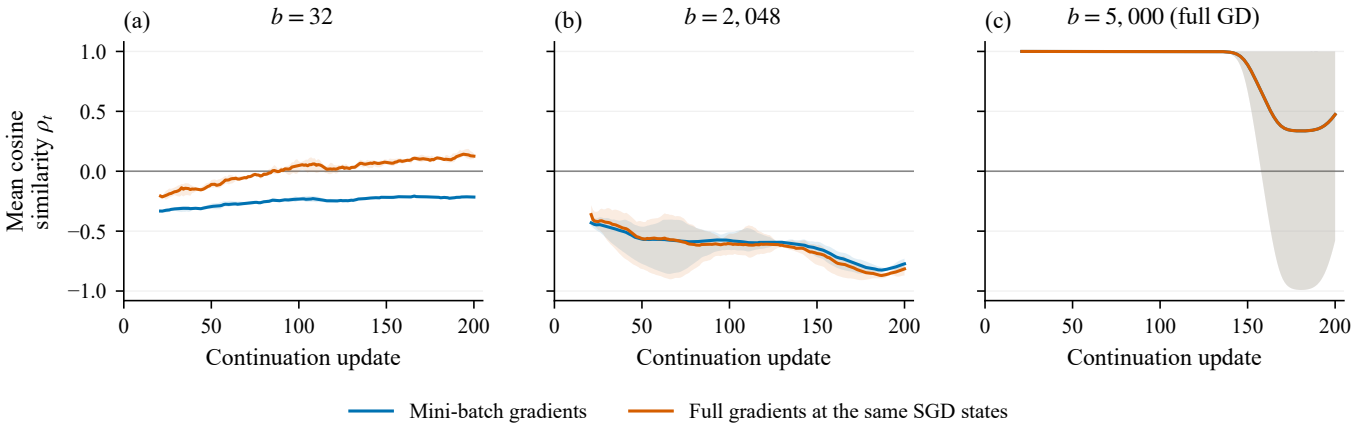


Fig. 5: Alignment during continuations from the 1,000-update state: smallest batch, largest stochastic batch and full-batch control. Curves show trailing 20-pair means, averaged over repetitions and initializations; shading is the range of initialization means. Full gradients are evaluated along each SGD path; the two curves coincide for full GD. The horizontal axis counts updates after the shared starting checkpoint.

Consequently, consecutive alignment, path-length retention and loss improvement must be read separately.

These observations explain why the sampled-gradient diagnostics retain information about actual SGD updates while requiring care as descriptions of full-gradient geometry. In particular, $\eta\widehat{\kappa}_t$ still contains the noise term in Eq. (16). The evidence covers one MLP objective, one learning rate and 200-update continuations; it does not establish a learning-rate dependence, a stability boundary, or a batch size that guarantees negligible noise.

VI. Directional Diagnostics under Adam

The geometric diagnostics developed for SGD can also characterize training under Adam, provided we distinguish the gradient information from the parameter motion produced by the optimizer. Raw gradients describe the instantaneous stochastic loss signal along the training trajectory; momentum averages this signal over time; and adaptive preconditioning determines how the accumulated information becomes a parameter update. We apply the same diagnostics to these successive quantities without transferring the SGD stability conditions to Adam.

A. Three Views of the Same Training Trajectory

For Adam [16], the first and second moments satisfy

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{24}$$

with zero initial moments. Let $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$ denote their bias-corrected values. The adaptive direction and parameter update are

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta_t d_t, \tag{25}$$

where squaring, square roots and division act coordinatewise. Here $\epsilon > 0$ is Adam's numerical constant, distinct from the sampling contribution ε_t in Section III. If gradients are clipped, the clipped vector replaces g_t in Eq. (24); raw-gradient diagnostics still use the unclipped gradient. AdamW additionally applies decoupled weight decay [28].

The norms, consecutive cosines and accumulation ratio from Section III can be computed for each of g_t, m_t and d_t. We identify the vector stream explicitly rather than redefine each diagnostic. Raw gradients reveal changes in the sampled loss signal; first moments show what temporal averaging retains; adaptive directions show the effect of coordinate scaling on that retained signal. All three are available within the optimizer, although measuring consecutive alignment requires retaining a previous vector, and cumulative ratios require a running vector sum. Their computation uses reductions rather than new backward passes; storage and communication costs still scale with the measured parameter scope.

An accumulation ratio computed from raw gradients or moments is not normalized parameter displacement under Adam. To recover that quantity, apply the same norm-of-sum over sum-of-norms construction to the actual parameter increments. Even for Adam without decay, these increments include η_t: varying the learning rate changes their relative weights. AdamW decay adds a separate displacement, and parameter-group learning rates or finite-precision updates can change the correspondence further. Positive scalar bias correction preserves consecutive momentum cosine, but generally changes cumulative accumulation because the scalar varies with time.

B. Raw Gradients as Observations of the Traversed Loss

Raw-gradient alignment remains informative under Adam because it records how the loss gradient changes along the optimizer's chosen path. The Taylor argument in Section III can be written using the actual preceding displacement:

$$\begin{aligned} g_t - g_{t-1} \approx H_{t-1}(\theta_{t-1} - \theta_{t-2}) \\ + \xi_t - \xi_{t-1}. \end{aligned} \tag{26}$$

This uses the same Hessian indexing and neglected remainder as Eq. (11). Under plain Adam, the displacement is $-\eta_{t-1} d_{t-1}$, rather than a scaled raw gradient. Consequently, the observed response combines local curvature, the direction selected by the optimizer and sampling variation. The SGD proxy in Eq. (16) remains computable as a normalized gradient-change statistic, but its directional-curvature interpretation does not carry over, even if sampling noise is small.

A return to positive raw-gradient cosine after reducing the learning rate therefore describes a change in the sampled gradient response. It does not by itself show that curvature decreased: shorter steps can change that response within the same local quadratic geometry. Nor does negative raw alignment imply that consecutive Adam updates oppose each other. That question must be answered from the transformed directions or actual parameter increments.

C. Momentum Growth, Rotation and Adaptive Scaling

The first-moment recurrence separates two effects of an incoming gradient. Its component along the stored moment can reinforce or oppose the current momentum, while its transverse component changes the direction. Increasing the magnitude of an aligned input can grow the moment without appreciable rotation. Increasing a transverse input relative to the stored moment can instead lower consecutive momentum cosine while that cosine remains positive. Thus, a decline from near one already indicates greater angular change; a sign reversal is not required.

Positive alignment between consecutive raw gradients does not guarantee that the latest gradient aligns with the longer history stored in m_{t-1}. Interpreting momentum rotation therefore benefits from also comparing the incoming optimizer gradient with the previous moment. This is a distinct comparison from adjacent raw-gradient cosine. With clipping, the magnitude entering the recurrence is the clipped magnitude: growth of the raw norm beyond the threshold need not produce corresponding moment growth. The signs and norms of the recorded streams alone do not identify that propagation mechanism.

Preconditioning adds another transformation. The same first moment can produce different update directions as coordinate weights change, and a large moment need not imply a large

TABLE II: Pythia-14M: median consecutive cosine over updates 612–711 and final validation loss. Raw, moment and adaptive columns refer to g_t, m_t and d_t, respectively. One initialization; rows are grouped by execution platform. Rates are rounded for display.

Rate	Raw	Moment	Adaptive	Val. loss
DGX Spark (GB10)				
10^{-4}	-0.334	0.867	0.978	6.232
10^{-3}	-0.432	0.916	0.942	4.764
10^{-2}	-0.103	0.403	0.854	6.636
RTX 4090				
2.15×10^{-4}	-0.434	0.573	0.972	5.601
4.64×10^{-4}	-0.495	0.762	0.958	5.238
2.15×10^{-3}	-0.310	0.956	0.912	4.410
4.64×10^{-3}	-0.361	0.852	0.854	4.318

adaptive step. Comparing momentum and adaptive-direction alignment reveals where their temporal behavior differs, but their numerical difference is not an additive measure of the preconditioner's effect. Likewise, a highly aligned adaptive sequence can still make poor objective progress. Loss measurements and controlled direction or step-size interventions are needed to connect these geometric descriptions to optimization outcomes.

D. Illustrations in Language-Model Training

We illustrate these distinctions with small Pythia-family decoder models [29], initialized from scratch and trained on MiniPile [30]. A seven-rate comparison uses one shared 14M initialization and data order (seed 1234), with linear warmup over the first 1% of training followed by a constant rate. Each run processes the same complete 1.491-billion-input-token cache in 712 updates, including the final partial batch. Full batches contain 1,024 packed sequences of length 2,048, accumulated in microbatches of four. AdamW uses $(\beta_1, \beta_2) = (0.9, 0.95)$, $\epsilon = 10^{-8}$, global gradient clipping at norm one, and decay 0.1 excluding biases and layer-normalization parameters. Computation is FP16 with FP32 parameters and moments; diagnostic reductions are FP64. Loss is averaged over supervised next-token targets. Validation covers all 500 held-out documents and 693,329 targets, including the packed tail, with token weighting.

Table II shows negative late raw-gradient alignment alongside positive first-moment and adaptive alignment at every tested rate. No recorded consecutive first-moment pair is negative in these seven runs, yet its late median falls to 0.403 at rate 10^{-2}. That run clips 517 of 712 updates and has late median raw norm 5.566, while the moment norm is only 0.097. These measurements illustrate the distinction between large raw gradients and the signal accumulated after clipping and averaging; they do not establish raw-norm growth as the cause of momentum rotation. Within the hardware-matched RTX 4090 runs, rate 4.64×10^{-3} has lower momentum alignment than 2.15×10^{-3} but better validation loss. The comparison is descriptive: it uses one initialization, and pooled differences across platforms cannot be attributed solely to learning rate.

A separate intervention branches 14M and 70M AdamW checkpoints at update 356 into three schedules: continue the original cosine decay, hold its current rate, or advance its decay clock twice as fast. Within each size, weights, optimizer state and future batches are matched; all branches continue through update 712 on RTX 5090 with the same recipe and validation coverage. The held rate is 5.1791×10^{-4}; original and accelerated decay reach the 10^{-4} floor at updates 575 and 466. Table III shows that faster decay makes raw alignment less negative, while held-rate continuations attain lower validation loss. Positive smoothed alignment also occurs earlier under accelerated decay, but later reverses in the 70M model. Changing the rate also changes AdamW's decay displacement, so these interventions identify schedule effects without isolating their mediation by step length or shrinkage.

TABLE III: Checkpoint-matched schedule intervention. Median raw cosine uses affected pairs 358–712; loss is final full-validation loss. Each size uses one initialization (seed 1235).

	14M		70M	
Schedule	Raw cosine	Val. loss	Raw cosine	Val. loss
Original decay	0.194	5.182	-0.109	4.056
Held rate	-0.358	5.067	-0.311	3.985
Faster decay	0.331	5.218	-0.006	4.104

E. From Characterization to Informed Scheduling

These observations motivate using the diagnostics to formulate and evaluate learning-rate interventions. Raw-gradient reversal with coherent momentum indicates that averaging is retaining a direction despite local variation. A simultaneous rise in raw norms, frequent clipping, declining momentum alignment and worsening loss supplies a different pattern that merits a step-size comparison. A change concentrated in adaptive directions calls for examining coordinate scaling and actual step lengths. These are interpretive patterns, not validated trigger thresholds.

An informed scheduling rule should therefore be evaluated by its subsequent loss improvement at a matched budget, rather than by whether it maximizes cosine similarity or displacement retention. The next test is to branch shared checkpoints into predefined rate interventions, record recent gradient, momentum and adaptive behavior, and compare later outcomes on held-out data across initializations. Such a test must establish whether the diagnostics add information beyond the current rate, training stage, clipping frequency and recent loss. The present extension supplies the measurement and interpretation framework for that test; it does not yet establish an improved adaptive schedule.

VII. Conclusion

The displacement formulation distinguishes the magnitude of individual updates, their local directional interaction and their cumulative accumulation. For plain SGD, the exact decomposition retains all temporal lags, while consecutive cosine provides a tractable local observation. The curvature and sampling relations clarify how that observation can change without determining either cumulative displacement or loss improvement. The full-batch and mini-batch experiments illustrate these distinctions: gradient reversal can coexist with

objective progress, and weaker reversal need not improve displacement retention or the attained loss.

Under Adam, the same measurements acquire different meanings at successive stages of the optimizer. Raw gradients describe the sampled loss response; first moments expose temporal averaging; adaptive directions incorporate coordinate scaling. Parameter displacement additionally depends on the applied learning rates and weight decay. The language-model comparisons show why positive momentum or adaptive alignment cannot serve as a universal performance criterion, and why restoring positive raw alignment through decay need not improve the final validation loss.

The practical contribution is a framework for interpreting these measurements together and using them to formulate controlled interventions. Developing a scheduler requires a further step: demonstrating that the diagnostics improve rate decisions beyond recent loss and known training settings, at matched budgets and across independent runs. That evaluation remains future work.

APPENDIX A
DERIVATION OF THE ALIGNMENT-BASED LOSS CHANGE

Consider the SGD update $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$. The change in the full objective along this step can be expressed exactly as

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad = -\eta \int_0^1 g_{t-1}^\top \nabla \mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds.\end{aligned} \tag{27}$$

Approximating the integral by the average of the full gradients at the two endpoints gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top \left(\nabla \mathcal{L}(\theta_{t-2}) + \nabla \mathcal{L}(\theta_{t-1})\right).\end{aligned} \tag{28}$$

This approximation is exact for quadratic objectives. For nonlinear objectives, the approximation error is of third order in the step length when the Hessian is locally Lipschitz continuous.

Using the stochastic gradient decomposition $g_t = \nabla \mathcal{L}(\theta_{t-1}) + \xi_t$ at both endpoints, we obtain

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2} g_{t-1}^\top (g_{t-1} + g_t - \xi_{t-1} - \xi_t) \\ &\quad = -\frac{\eta}{2} \left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t).\end{aligned} \tag{29}$$

Substituting $g_{t-1}^\top g_t = \|g_{t-1}\| \|g_t\| \rho_t$ and factoring the preceding squared gradient norm yields

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\frac{\eta}{2} \|g_{t-1}\|^2 \left(1 + \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}\right) \\ &\qquad + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t),\end{aligned} \tag{30}$$

which recovers Eq. (22).

To connect this expression with directional curvature, recall from Eq. (15) that

$$\rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx 1 - \eta \kappa_t + \varepsilon_t. \tag{31}$$

Substituting this relation and using the definition of ε_t in Eq. (14) gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\eta \|g_{t-1}\|^2 \left(1 - \frac{\eta \kappa_t}{2}\right) + \eta \xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{32}$$

Thus, the curvature-based and alignment-based expressions describe the same local loss change to second order. Positive directional curvature counteracts the first-order loss decrease, while its effect is already incorporated into the weighted alignment between consecutive gradients. The noise contributions account for the difference between the observed mini-batch gradients and the full gradients.

APPENDIX B
DERIVATION OF THE LOCAL LOSS CHANGES

We use the same indexing as Section III: g_{t-1} is evaluated at θ_{t-2}, g_t at θ_{t-1}, and $H_{t-1} = \nabla^2 \mathcal{L}(\theta_{t-2})$. The loss is assumed twice continuously differentiable along the displacements considered below. The relations retain second-order terms and omit the Taylor remainders; they are exact when the objective is quadratic.

A. One Update and Directional Curvature

For $\theta_{t-1} - \theta_{t-2} = -\eta g_{t-1}$, Taylor expansion gives

$$\begin{aligned}&\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \\ &\quad \approx -\eta \nabla \mathcal{L}(\theta_{t-2})^\top g_{t-1} + \frac{\eta^2}{2} g_{t-1}^\top H_{t-1} g_{t-1}.\end{aligned} \tag{33}$$

Substituting $\nabla \mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $g_{t-1}^\top H_{t-1} g_{t-1} = \kappa_t \|g_{t-1}\|^2$ yields Eq. (22) for nonzero g_{t-1}.

The positive term $\eta \xi_{t-1}^\top g_{t-1}$ in that expression cannot be interpreted separately as the net effect of noise. Conditional on the parameters before the batch is drawn, unbiased sampling gives

$$\begin{aligned}\mathbb{E}[\|g_{t-1}\|^2 \mid \theta_{t-2}] &= \|\nabla \mathcal{L}(\theta_{t-2})\|^2 \\ &\quad + \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}], \\ \mathbb{E}[\xi_{t-1}^\top g_{t-1} \mid \theta_{t-2}] &= \mathbb{E}[\|\xi_{t-1}\|^2 \mid \theta_{t-2}].\end{aligned} \tag{34}$$

Thus, the contributions from noise variance cancel between the first-order terms when taking their conditional expectation. With finite moments for the retained terms, the expected loss change in the quadratic approximation is

$$\begin{aligned}&\mathbb{E}[\mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2}) \mid \theta_{t-2}] \\ &\quad \approx -\eta \|\nabla \mathcal{L}(\theta_{t-2})\|^2 \\ &\qquad + \frac{\eta^2}{2} \mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{35}$$

Sampling still contributes to the curvature-weighted second-order term. This conditional calculation does not remove that contribution or the Taylor remainder from a nonlinear trajectory.

B. Two Updates and Consecutive Alignment

The displacement over two SGD updates is exactly

$$\theta_t - \theta_{t-2} = -\eta(g_{t-1} + g_t).$$

Expanding around the starting parameters θ_{t-2} gives

$$\begin{aligned}\mathcal{L}(\theta_t) - \mathcal{L}(\theta_{t-2}) \\ \approx -\eta \nabla \mathcal{L}(\theta_{t-2})^\top (g_{t-1} + g_t) \\ + \frac{\eta^2}{2}(g_{t-1} + g_t)^\top H_{t-1}(g_{t-1} + g_t).\end{aligned} \tag{36}$$

The first-order projection can be written as

$$\begin{aligned}\nabla \mathcal{L}(\theta_{t-2})^\top (g_{t-1} + g_t) \\ = \|g_{t-1}\|^2 + g_{t-1}^\top g_t - \xi_{t-1}^\top (g_{t-1} + g_t).\end{aligned} \tag{37}$$

Finally, $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ yields a form of Eq. (36) using consecutive-gradient cosine. The same starting Hessian appears throughout this expansion. For a nonlinear objective, its remainder is governed by the combined displacement; a small cosine alone does not control that approximation error.

Appendix C
Measurement Details and Additional Full-Batch Diagnostics

A. Objectives and Measurement

The MLP uses coordinate-wise standardization and ResNet uses channel-wise standardization, both based on the complete CIFAR-10 training split. ResNet's objective averages the same 50 fixed groups of 1,000 images, with batch normalization performed separately within each group. Accumulating all group gradients gives the gradient of this fixed objective; physical batching introduces no random sampling.

For the Transformer, the corpus is arranged into 20 fixed streams and causal chunks of at most 35 tokens. Eight stream-tail tokens and the 20 initial context tokens have no training target; short chunks retain their valid targets, and padding is excluded from the loss. The resulting 59,680 sequences contain 2,088,600 training targets. The 33,278-word vocabulary follows the first-occurrence dictionary over training, validation and test text in the PyTorch word-language-model example, revision `13acec6d7c78` [31]. The same pinned example supplies the Transformer implementation. These comparisons concern training dynamics rather than held-out generalization.

Physical accumulation uses 1,000-image chunks for the MLP, the fixed 1,000-image groups for ResNet, and 64-sequence chunks for the Transformer. The complete objective is accumulated before each parameter update. Training parameters and gradients are FP32, TF32 is disabled, and diagnostic reductions are FP64. The nine conditions are paired learning-rate comparisons within three model cases, not nine independent initializations or a controlled comparison of architectures. The gradient diagnostics require vector reductions and retained gradient information rather than online eigendecomposition; their runtime and memory overhead have not been isolated by a controlled benchmark.

The first adjacent pair is omitted without a denominator offset. The complete 50-observation trailing means aid visualization and do not define regime-onset times. Endpoint losses and R_{3000} are unsmoothed.

B. Unsmoothed Alignment and Temporal Accounting

Figure 6 supplies the unsmoothed consecutive cosines underlying Fig. 2(a)–(c). Complete trailing means suppress short transients and shift the displayed transition relative to the first negative pair. In particular, the MLP at $\eta = 0.04$ reverses near initialization and then returns to positive alignment before its later negative phase. These raw traces distinguish such a transient from a persistent pattern without introducing a stability threshold or discarding early observations.

For the complete prefix of T gradients, define the traveled distance and the temporal contributions as

$$\begin{aligned}P_T &:= \eta \sum_{t=1}^{T} \|g_t\|, \\ I_T^{(1)} &:= 2 \sum_{t=2}^{T} g_{t-1}^\top g_t, \\ I_T^{(>1)} &:= 2 \sum_{\substack{1 \le u < t \le T \\ t-u > 1}} g_u^\top g_t, \\ I_T &:= I_T^{(1)} + I_T^{(>1)} = \Big\| \sum_{t=1}^{T} g_t \Big\|^2 - \sum_{t=1}^{T} \|g_t\|^2.\end{aligned} \tag{38}$$

Table IV reports these quantities at $T = 3000$. The interaction columns use raw gradients and include the factor of two; they are not multiplied by η^2. The total is positive in all nine conditions, whereas the adjacent contribution is negative in seven. This endpoint result does not assert positivity at every prefix or for every individual pair. Comparisons of magnitudes are within each model case and its fixed parameterization.

C. Retrospective Loss Approximation

For each gradient pair indexed by $t = 2, \ldots, 3000$, let $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. We compare this observed change with $\widehat{\delta\mathcal{L}}_{t-1}$ from Eq. (23). The normalized absolute discrepancy is

$$e_t := \frac{|\delta\mathcal{L}_{t-1} - \widehat{\delta\mathcal{L}}_{t-1}|}{\eta \|g_{t-1}\|^2}. \tag{39}$$

The denominator is the magnitude of the first-order full-GD descent term. It is nonzero for every measured pair. Table V reports the median over all 2,999 pairs and the count of opposite signs. The final update is excluded because its subsequent gradient was not recorded. All inputs are unsmoothed; the table uses no seed aggregation.

For the threshold sensitivity columns, define

$$h_t(\tau) := \tau \max\{|\mathcal{L}(\theta_{t-2})|, |\mathcal{L}(\theta_{t-1})|\}. \tag{40}$$

A transition is eligible only when both $|\delta\mathcal{L}_{t-1}| > h_t(\tau)$ and $|\widehat{\delta\mathcal{L}}_{t-1}| > h_t(\tau)$. Each cell prints opposite-sign transitions divided by its eligible count; the median discrepancy remains computed over all pairs. These thresholds are descriptive sensitivity choices, not calibrated FP32 error bounds. Serial transitions are not independent initialization replicates, and the comparison is retrospective because it uses the subsequent gradient. The discrepancy does not isolate finite-step, nonsmooth or numerical effects.

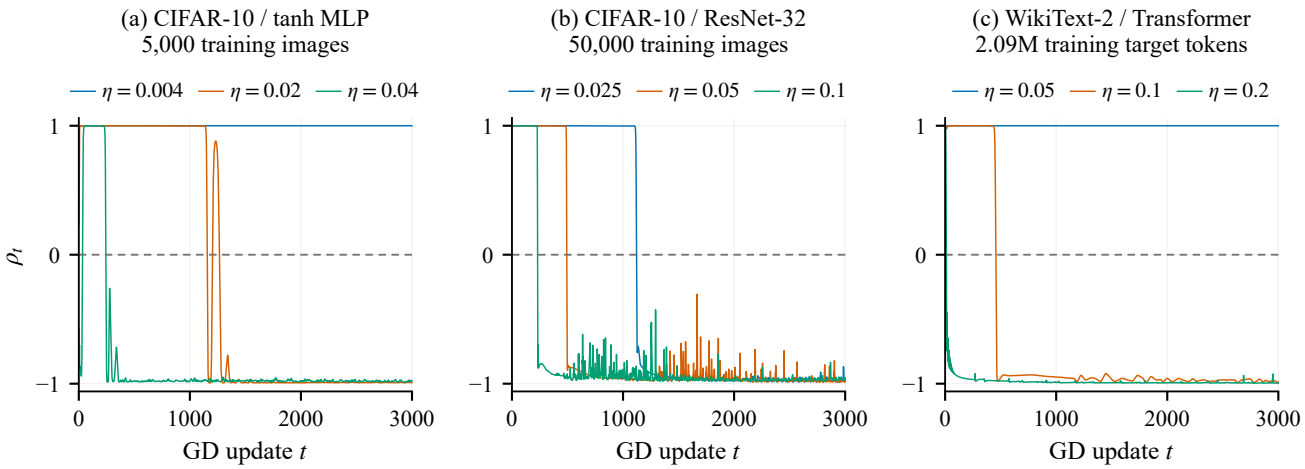


Fig. 6: Unsmoothed consecutive-gradient cosine for the nine full-GD conditions in Section IV, with 2,999 defined pairs per condition. Columns show (a) MLP, (b) ResNet-32 and (c) Transformer. The dashed horizontal reference is zero. Blue, orange and green identify increasing learning rates within each case, as given in the legends. Every curve uses seed 0 and covers the same 3,000-update horizon; the first undefined pair is omitted and there is no temporal or seed averaging. Coincident positive curves can overlap exactly.

D. Spectral Context

We measure the largest algebraic Hessian eigenvalue at states $0, 100, \ldots, 3000$ using matrix-free Hessian–vector products and a checked relative eigenpair residual of at most 0.005. The Hessian objective covers all 5,000 MLP training examples, but only the first five fixed ResNet groups and the first 2,500 Transformer sequences. Subset measurements follow the computational motivation of [13]; they are not full-objective sharpness for the larger models. Reused MLP trajectories requested two largest eigenvalues and the other runs requested one; we compare only the largest value. Table VI retains all nine final measurements.

Maximum pointwise eigenvalue concerns the most curved local direction, whereas $\widehat{\kappa}_t$ concerns realized gradient change across a step. Objective coverage and a finite-step Taylor remainder also affect their comparison. Final spectral values and the late means in Table I use different temporal reductions. Their agreement or disagreement therefore does not isolate estimator accuracy or the cause of a discrepancy. A state-matched local comparison would pair the Hessian at state s with the expansion for gradient index $t = s + 2$; the spectrum at state 3000 has no matching recorded gradient pair.

TABLE IV: Complete-prefix temporal accounting at $T = 3000$, with the quantities defined in Eq. (38). P_{3000} is path length; the last three columns are raw-gradient interactions. All nine conditions use one shared initialization per case. Values are rounded only for display, so displayed components may not sum exactly to the displayed total.

Model	η	P_{3000}	$I^{(1)}_{3000}$	$I^{(>1)}_{3000}$	I_{3000}
MLP	0.004	1.1976	1.270×10^{2}	5.634×10^{4}	5.647×10^{4}
MLP	0.02	64.3608	-1.217×10^{4}	3.063×10^{4}	1.846×10^{4}
MLP	0.04	185.5420	-1.662×10^{4}	2.225×10^{4}	5.630×10^{3}
ResNet-32	0.025	41.1785	-2.528×10^{3}	5.325×10^{4}	5.072×10^{4}
ResNet-32	0.05	146.3482	-7.380×10^{3}	3.603×10^{4}	2.865×10^{4}
ResNet-32	0.1	372.7896	-1.076×10^{4}	2.451×10^{4}	1.375×10^{4}
Transformer	0.05	15.6687	1.629×10^{2}	5.100×10^{4}	5.116×10^{4}
Transformer	0.1	111.8465	-7.767×10^{2}	2.464×10^{4}	2.387×10^{4}
Transformer	0.2	405.8404	-2.682×10^{3}	1.336×10^{4}	1.067×10^{4}

TABLE V: Retrospective check of the local loss approximation for all nine conditions. The unthresholded count and median e_t use 2,999 transitions per condition. Each threshold cell gives opposite signs / eligible transitions; eligibility requires both change magnitudes to exceed $h_t(\tau)$. These descriptive temporal counts are not independent-sample predictive accuracies.

				Opposite signs / eligible transitions at threshold τ		
Model	η	No threshold	Median e_t	10^{-7}	10^{-6}	10^{-5}
MLP	0.004	0/2999	2.905×10^{-4}	0/2999	0/2999	0/2999
MLP	0.02	145/2999	6.146×10^{-4}	145/2998	145/2998	141/2989
MLP	0.04	281/2999	7.343×10^{-3}	281/2999	281/2998	280/2994
ResNet-32	0.025	595/2999	1.030×10^{-2}	595/2999	592/2991	570/2934
ResNet-32	0.05	694/2999	2.347×10^{-2}	694/2999	692/2997	682/2978
ResNet-32	0.1	814/2999	2.324×10^{-2}	813/2997	813/2997	809/2989
Transformer	0.05	0/2999	2.381×10^{-5}	0/2999	0/2999	0/2999
Transformer	0.1	542/2999	1.398×10^{-2}	541/2997	534/2977	436/2703
Transformer	0.2	1325/2999	6.323×10^{-2}	1324/2996	1319/2989	1239/2837

TABLE VI: Final spectral context at parameter state 3000. Only the MLP measurement covers its complete training objective. ResNet and Transformer use fixed subsets of the stated size; residual-checked eigenpairs do not establish subset representativeness. All conditions use seed 0.

Model	η	Hessian objective coverage	$\eta\lambda_{\max,3000}$
MLP	0.004	Full 5,000 images	0.3168
MLP	0.02	Full 5,000 images	2.2157
MLP	0.04	Full 5,000 images	2.3259
ResNet-32	0.025	Fixed 5,000-image subset	2.8176
ResNet-32	0.05	Fixed 5,000-image subset	2.6599
ResNet-32	0.1	Fixed 5,000-image subset	3.2488
Transformer	0.05	Fixed 2,500-sequence subset	1.7229
Transformer	0.1	Fixed 2,500-sequence subset	2.2752
Transformer	0.2	Fixed 2,500-sequence subset	3.1202

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